

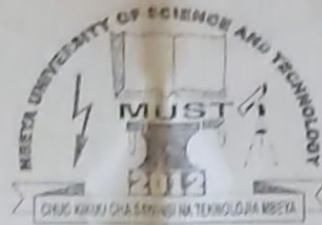
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# MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CONTINUOUS ASSESSMENT TEST-I APRIL 2024

ET 8208: DATA STRUCTURE AND ALGORITHMS ANALYSIS.

TIME: 1:30hrs

24<sup>th</sup> April 2024

Answer ALL questions. [3 marks each]

## Question One

- a) Define data structure, use a diagram to describe its classification, giving an example for each.

## Question Two

- a) Give a brief discussion on any five characteristics of algorithms.

## Question Three

- a) Why algorithm analysis is important? Explain briefly how it can be done.

## Question Four

- a) Describe how the run time of a program can be calculated.

## Question Five

- a) Using a diagram, explain how array elements are accessed and stored.